

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456944.78449 +/- 0.00077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1154 +/- 0.0058
Transit depth (Rp/Rs)^2	1.33 +/- 0.13 [%]
Orbital Inclination (inc)	89.86 +/- 0.58
Airmass coefficient 1 (a1)	1.1344 +/- 0.0068
Airmass coefficient 2 (a2)	-0.1194 +/- 0.0033
Scatter in the residuals of the lightcurve fit is	0.6 %
Best Comparison Star	None
Optimal Aperture	4.05
Optimal Annulus	8.87
Transit Duration (day)	0.1051 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1154 ± 0.0058 vs 0.131 (deviation 2.45σ)
Duration fit vs published	0.1051 d vs 0.1075 d (deviation 1.99σ)
Mid-time O–C	2.6 min
Depth SNR	18.1
Residual scatter	0.6 %

Light curve

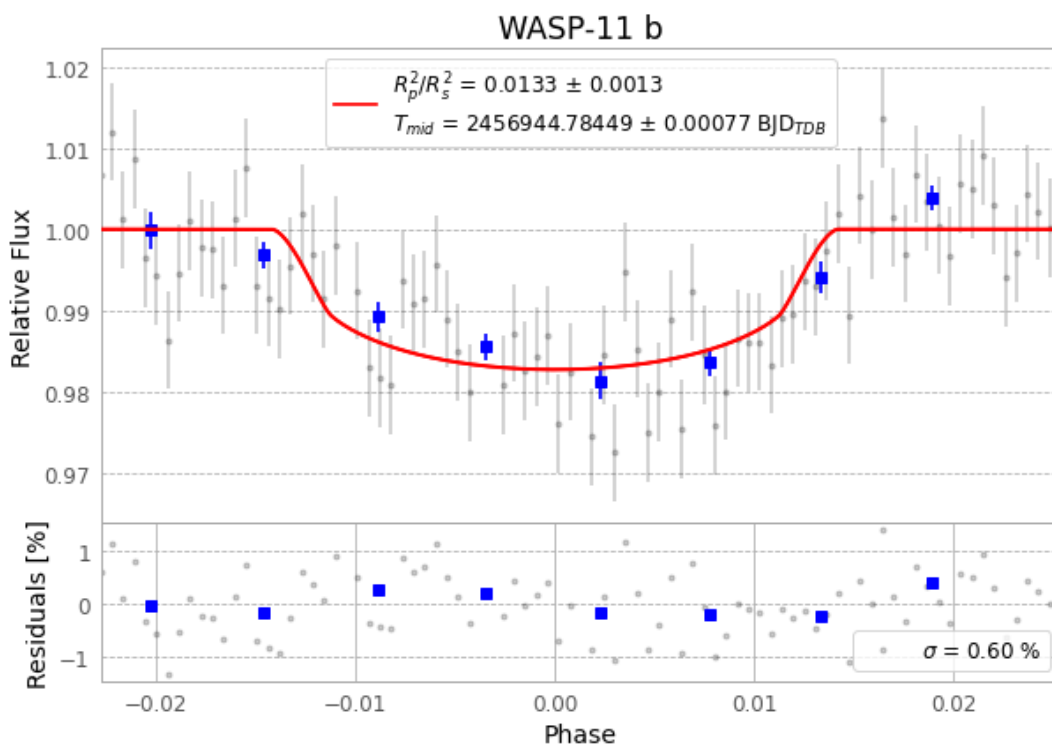


Figure 1 — FinalLightCurve_WASP-11 b_2014-10-13.png (SHA-256 5c50673c9aeb24c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [262, 348], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f07134a32c53fb2a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), HATP-10141014043912.FITS → HATP-10141014085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-141014.log (9e05258e3306...), FinalLightCurve_WASP-11 b_2014-10-13.png (5c50673c9aeb...), FinalLightCurve_WASP-11 b_2014-10-13.csv (ae22bc1aa2bd...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:33Z.